

# Tensile property analysis of carbon/glass hybrid fibers-reinforced graphene-modified polymer bars

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## ABSTRACT

The application of single-type fiber-reinforced polymer (single-FRP) bar in civil engineering is restricted due to the brittle failure and low tensile modulus. In this work, the carbon/glass hybrid fibers-reinforced polymer (C/G-HFRP) bars were designed to realize tensile ductility and improve tensile modulus. Also, graphene nanosheets (GNPs) were used to modify the fracture toughness of polymer matrix between fibers, which can resist the energy release rate during the failure of carbon fibers and the crack initiation and development inside the modified C/G-HFRP bars. The experimental results indicated that GNPs increased the cooperative ability between fibers and reduced the separation degree of glass fibers after the failure of carbon fibers. The tensile strength and strain of the modified C/G-HFRP bars at the initial failure of carbon fibers were averagely improved by 6.9% and 4.3% relative to those of the unmodified ones, while their ultimate tensile strength and strain were averagely increased by 7.1% and 7.1%, respectively. Meanwhile, four evaluation indicators were proposed to reflect the superior performance of the modified C/G-HFRP bars. Additionally, the experiments verified that the contribution of carbon fibers to the secondary stiffness at the strain hardening stage in the tensile stress-strain response of the modified C/G-HFRP bars can be omitted.

## 1. Introduction

Corrosion of steel bars normally reduces the stiffness and load-bearing capacity of concrete structures. The repair or reconstruction of concrete structures caused by corrosion of steel bars require significant resources and economic costs [1,2]. Currently, a novel alternative or remedy is urgently needed to solve the corrosion problem of steel bars [3–6]. Comparatively, single-type fiber-reinforced polymer (single-FRP) bar with corrosion resistance is an attractive substitute for the steel bars [7,8]. The single-FRP bar consists of single-type continuous fibers and polymer matrix, of which the continuous fibers primarily undertake load and provide stiffness, and the polymer matrix coheres continuous fibers together [9]. Through the pultrusion-winding process, the continuous fibers impregnated with the polymer matrix are squeezed and then thermally cured to produce various single-FRP bars.

Although the single-FRP bar has a high-specific strength and stiffness, resistant-corrosion, and impact resistance, the widespread application of single-FRP bar in civil engineering is still restricted due to its brittle failure and low tensile modulus [10]. The brittle failure of single-FRP bar results from the catastrophic destruction of single-type

continuous fibers under tensile load and the brittle fracture of polymer matrix. Hybrid fibers with two or more stiffnesses, which serve as the reinforced phase of polymer matrix, can make them break in stages under tensile load to realize the plastic failure of FRP bar. Meanwhile, the low tensile modulus of single-FRP bar can also be improved by adding high-stiffness fibers [11]. This novel FRP bar is called hybrid fibers-reinforced polymer (hybrid-FRP) bar [10–12]. Additionally, the polymer matrix of FRP bar is usually served by epoxy, vinyl resin, or unsaturated polyester resin, which behave the intrinsic defects of brittle failure, poor crack resistance, and lack of impact resistance under applied load [13,14]. To improve the crack resistance of these resins, nanoparticles are normally designated as fillers to modify their fracture toughness. However, the incorporating content of nanoparticle is critical to balance the impregnation effect of modified resin on fibers and the performance improvement of modified resin. Therefore, the selection of nanoparticle is cautious to toughen the polymer matrix of hybrid-FRP bar.

Breakage in stages of hybrid fibers has been introduced to solve the brittle failure of single-FRP bar. Belarbi et al. [15] pointed out that the breakage in stages of hybrid fibers with two or more stiffnesses realized

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the plastic deformation of hybrid-FRP bar under tensile load. Bakis et al., [16] Ali et al. [17] and Jones et al. [18] reported that the volume fraction and distribution configuration of high-stiffness fibers affected the tensile ductility of hybrid-FRP bar. The results indicated that the lower the volume fraction of high-stiffness fibers and the more dispersed the distribution configuration, the higher the ductility of hybrid-FRP bar under tensile load. To avoid simultaneous breakage of hybrid fibers under tensile load, Gao et al. [19] provided the calculation method of the critical ratio of high-stiffness to low-stiffness fiber volume fractions. Moreover, Gao et al. [20] imitated the strength grade of steel bars to design and verify experimentally the feasibility of hybrid-FRP bars with different strength grades. Furthermore, a five-stage tensile constitutive model was also proposed by Gao et al. for the hybrid-FRP bars with two types of fibers [21].

Additionally, studies on the modification of polymer matrix by incorporating nanoparticles have been extensively reported [22–29]. The conclusions indicated that the incorporating content of ordinary nanoparticles normally occupied for 5 to 20 wt% when achieving a significant improvement. Szmigiera et al. [30] incorporated 25 to 30 wt % nanosilica into the epoxy matrix of hybrid-FRP bar. The result showed that the mechanical properties of modified hybrid-FRP bars were 3% lower than those of the samples before modification, resulting from the uneven dispersion of nanosilica particles. An increment of 18% in the tensile strength was reported by Su et al. [31] for basalt-FRP bar, of which the polymer matrix was modified by adding 3 wt% nanosilica particles. Also, Su et al. [32] used the multi-wall carbon nanotubes (MCNTs) to toughen the polymer matrices of basalt-, carbon-, and glass-FRP bars. The findings claimed that the addition of 0.3 wt% MCNTs effectively improved their mechanical properties. It is noted that the addition of 5 to 20 wt% ordinary nanoparticles inevitably increases the viscosity of polymer matrix to bring about a poor impregnation effect on fibers. Hence, a low incorporating nanoparticle should be designated to toughen the polymer matrix of hybrid-FRP bar. With the prevalence of graphene nanosheets (GNPs), the excellent cost performance makes it affordable for the public [33,34]. Therefore, GNPs with a large-specific area may be an optimal choice to toughen the polymer matrix of hybrid-FRP bars [35,36]. Rafiee et al. [37] tested the mechanical properties of the epoxy modified by adding 0.1 wt% GNPs or CNTs, respectively. The results demonstrated that the mechanical properties of GNPs-modified epoxy were superior to those of CNTs-modified one. The similar conclusion was also declared by Wang et al. [38].

In this work, carbon and glass fibers were mingled to solve the brittle failure and low tensile modulus of glass-FRP bar. Also, GNPs were employed to toughen the polymer matrix of carbon/glass hybrid-FRP (C/G-HFRP) bar. The effects of the ratio of carbon to glass fiber volume fractions, distribution configuration of carbon fibers, total volume fraction of fibers, and modification of polymer matrix on the tensile properties of C/G-HFRP bar were investigated. X-ray diffraction (XRD) and scanning electron microscopy (SEM) were used to identify the physical phases and toughening mechanisms of the modified polymer matrix, respectively. The evaluation indicators, including the hybrid effect, ductile strain, stress drop rate, and strength ratio, were adopted to reflect the superior performance of modified C/G-HFRP bars. Finally, the contribution of carbon fibers to the secondary stiffness of the strain hardening stage in the tensile stress–strain responses of C/G-HFRP bars was experimentally analyzed.

## 2. Testing program

### 2.1. Design of ductility

To avoid simultaneous breakage of carbon and glass fibers under tensile load, the breakage theory in stages of fibrous composite was proposed to determine the critical ratio  $(V_C/V_G)_{Critical}$  of carbon to glass fiber volume fractions in the ductile C/G-HFRP bar as follows:

$$\left(\frac{V_C}{V_G}\right)_{Critical} = \frac{E_G(\varepsilon_G - \varepsilon_C)}{E_C \varepsilon_C} \quad (1)$$

where  $V_C$  and  $V_G$  signify the volume fractions of carbon and glass fibers in C/G-HFRP bars, respectively;  $E_C$  and  $E_G$  denote the tensile moduli of carbon and glass fibers, respectively;  $\varepsilon_C$  and  $\varepsilon_G$  are the ultimate tensile strains of carbon and glass fibers, respectively.

The tensile ductility of composite bars can be realized when  $(V_C/V_G)_{Design}$  in C/G-HFRP bar is not larger than  $(V_C/V_G)_{Critical}$  as follows:

$$\left(\frac{V_C}{V_G}\right)_{Design} \leq \left(\frac{V_C}{V_G}\right)_{Critical} \quad (2)$$

### 2.2. Constituent materials

The reinforced phase of C/G-HFRP bars consisted of carbon and glass fibers, and the bisphenol A epoxy vinyl resin was chosen as their matrix phase. Multilayer GNPs were employed to toughen the bisphenol A epoxy vinyl resin. Table 1 lists the performance parameters of the fibers and resin provided by manufacturers, except that the tensile modulus and fracture toughness of the epoxy vinyl before and after modification were obtained based on the experiments by authors [39].

### 2.3. Test schemes

By substituting the material parameters in Table 1 into Eq. (1) and taking  $\varepsilon_G$  as 2.5%,  $(V_C/V_G)_{Critical}$  was determined to be 1:4.30. According to Eq. (2), four  $V_C/V_G$  from 1:10 to 1:4 were chosen to prepare different C/G-HFRP bars with ductility. On the basis of previous research [39], GNPs with a 0.3 wt% incorporating content was evenly dispersed into the polymer matrix of C/G-HFRP bars.

Table 2 gives the characteristic parameters of C/G-HFRP bars before and after modification using GNPs, where  $V_C^D/V_G^D$  and  $V_C^A/V_G^A$  denote design and actual values of  $V_C/V_G$  in C/G-HFRP bars, respectively;  $V_f^D$  and  $V_f^A$  are design and actual values of volume fraction of total fibers ( $V_f$ ) in C/G-HFRP bars, respectively. Based on the values designated in Table 2, incomplete bundles of the carbon and glass fibers with an open cross-sectional area were included; in fact, the incomplete fiber bundles cannot be realized, because the carbon and glass fiber bundles have a closed cross-sectional area when they are purchased from manufacturers. Hence, complete bundles of the carbon and glass fibers were adopted based on the rounding-off rule, which generated the difference between the design values and actual results.

As shown in Table 2, according to the distribution configuration of carbon fibers, the C/G-HFRP bars can be separated into two categories, including centralization and dispersion. In the centralized configuration, all carbon fibers were placed at the core, while the carbon fibers were regularly dispersed throughout the cross-sectional area in the dispersed configuration. In this work,  $V_f^D$  of 60% was employed for most of the C/

**Table 1**  
Material performance.

| Materials (Manufacturer) | Tensile strength (MPa) | Tensile modulus (GPa) | Tensile strain (%) | Fracture toughness (MPa·m <sup>1/2</sup> ) |
|--------------------------|------------------------|-----------------------|--------------------|--------------------------------------------|
| Carbon fibers (Toray)    | 4400                   | 377                   | 1.2                | –                                          |
| Glass fibers (Jushi)     | 2740                   | 81                    | 2.0 ~ 3.0          | –                                          |
| Epoxy vinyl (Wanqing)    | 70                     | 3                     | 5.0                | 1.044 ± 0.073                              |
| Modified epoxy vinyl     |                        | 3.39 ± 0.203          |                    | 1.275 ± 0.154                              |
| GNPs (Chunliang)         | Diameter (μm)          | Tensile modulus (GPa) | Thickness (nm)     | Number of plies                            |
|                          | 5                      | 1000                  | 5 ~ 7              | 3 ~ 5                                      |

**Table 2**

Characteristic parameters of C/G-HFRP bars.

| Sample code | $V_C^D/V_G^D$ | $V_C^A/V_G^A$ | Distribution configuration | $V_f^D$ | $V_f^A$ | Content of GNP (wt %) |
|-------------|---------------|---------------|----------------------------|---------|---------|-----------------------|
| GC0G1N60    | 0:1           | 0:1.00        | None                       | 0.6     | 0.613   | 0.3                   |
| HC1G10C60   | 1:10          | 1:11.40       | Centralized                | 0.6     | 0.627   | 0.3                   |
| HC1G8C60    | 1:8           | 1:8.20        |                            |         | 0.606   |                       |
| HC1G6C60    | 1:6           | 1:6.24        |                            |         | 0.585   |                       |
| HC1G4C60    | 1:4           | 1:4.34        |                            |         | 0.621   |                       |
| HC1G10D60   | 1:10          | 1:11.40       | Dispersed                  | 0.6     | 0.627   | 0.3                   |
| HC1G8D60    | 1:8           | 1:8.20        |                            |         | 0.606   |                       |
| HC1G6D60    | 1:6           | 1:6.24        |                            |         | 0.585   |                       |
| HC1G4D60    | 1:4           | 1:4.34        |                            |         | 0.621   |                       |
| HC1G6C50    | 1:6           | 1:6.11        | Centralized                | 0.5     | 0.503   | 0.3                   |
| HC1G6C70    | 1:6           | 1:5.70        |                            | 0.7     | 0.678   |                       |

G-HFRP bars, while  $V_f^D$  of 50% and 70% were also adopted corresponding to a  $V_C^D/V_G^D$  of 1:6. Additionally, the C/G-HFRP bars without modification using GNPs were also prepared for the sake of comparison. The coding notions in Table 2 can be explained as follows: the letters denote successively FRP bar type, carbon fibers, glass fibers, and distribution configuration of carbon fibers; the numbers signify  $V_C^D/V_G^D$  and  $V_f^D$ , respectively. For example, sample HC1G4C60 stands for a hybrid-FRP bar, which has a  $V_C^D/V_G^D$  of 1:4, a  $V_f^D$  of 60%, and a centralized

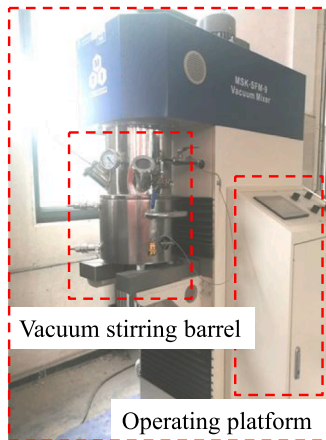
carbon fiber configuration.

#### 2.4. Modification of matrix

GNPs with 0.3 wt% incorporating content were evenly dispersed into the polymer matrix of C/G-HFRP bars by a high-speed shear vacuum mixer (MSK-SFM-9-5L), of which the photograph and the operating parameters are presented in Fig. 1(a). The stirring regime for uniformly dispersing GNPs into matrix is shown in Fig. 1(b). The experiment results of tensile modulus and fracture toughness are exhibited in Table 1 for the polymer matrix after modification using GNPs, which are improved by 13.0% and 22.1% relative to those without modification, respectively.

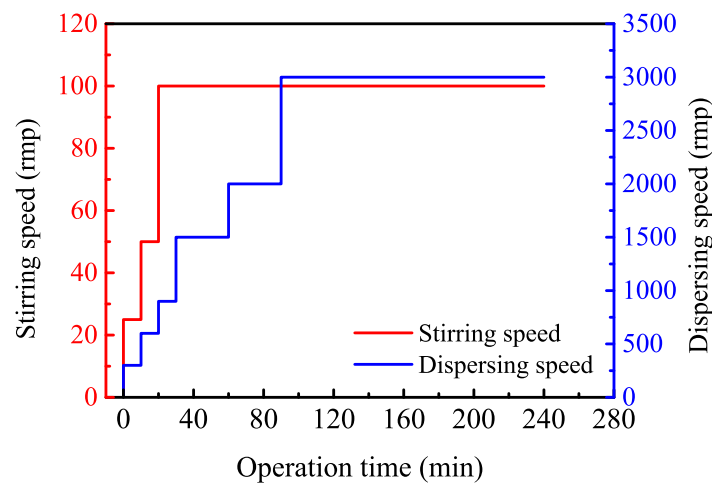
#### 2.5. Production of C/G-HFRP bars

Fig. 2(a) exhibits photograph and diagram of the pultrusion/winding equipment used, consisting of (A) yarn frame, (B) impregnation device, (C) pre-form mold, (D) front pre-heating box, (E) secondary extrusion mold, (F) winding device, (G) post pre-heating box, (H) gel box, (I) curing box, (J) cooling box, and (K) traction device. As shown in Fig. 2 (b), the epoxy vinyl resin modified using GNPs is firstly poured into the impregnation device; the carbon and glass fiber bundles calculated are pulled out from the yarn frame to pass through the yarn guide hole and



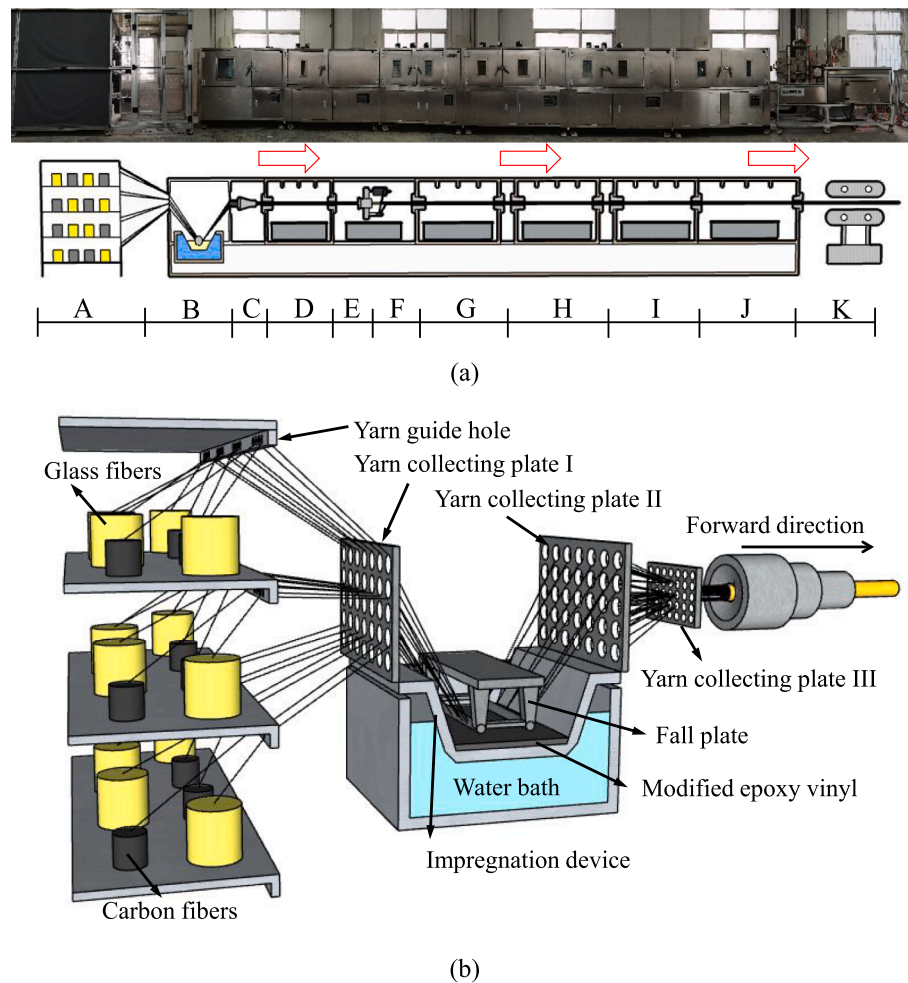
| Operating parameters (MSK-SFM-9-5L) |                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------|
| Power supply                        | Voltage: 220V (single-phase); Frequency: 50Hz; Power: 4.5 kW                          |
| Stirring pattern                    | Revolution & Rotation (Two stirring shafts & One dispersing shaft & One wall scraper) |
| Applicable size                     | Medium density: $\leq 2$ kg/L<br>Viscosity: $\leq 6 \times 10^5$ mPa·s                |
| Rotating speed                      | Revolution: $\leq 70$ rpm; Rotation: $\leq 112$ rpm;<br>Dispersion: $\leq 6100$ rpm   |
| Vacuum                              | Vacuum degree: $-0.098$ MPa                                                           |
| Stirring time                       | 0~999 min                                                                             |
| Volume                              | Effective volume: $\leq 5$ L; Design volume: 8L                                       |

(a)



(b)

**Fig. 1.** Matrix modification: (a) high-speed shear vacuum mixer and operation parameters and (b) stirring regime.

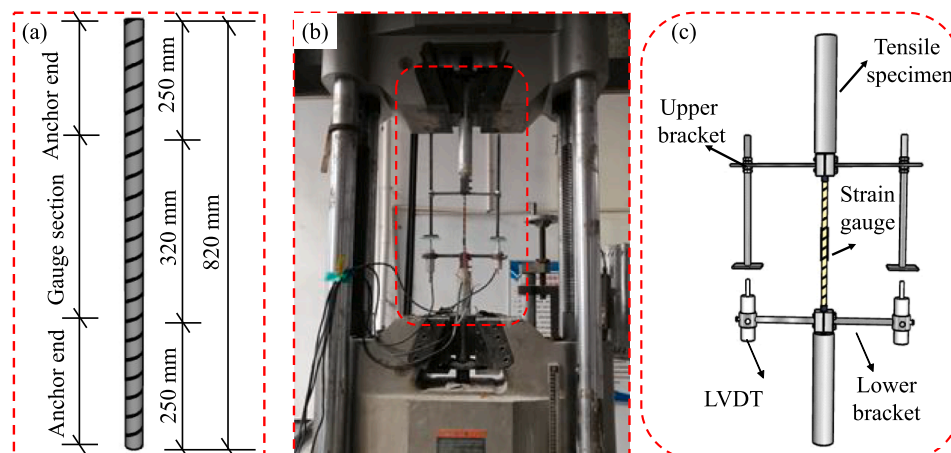


**Fig. 2.** Pultrusion/winding equipment: (a) photograph and sketch and (b) details of fiber impregnation.

collecting plates, and then immersed into the impregnation device. Next, under the action of traction device, the carbon and glass fiber bundles impregnated with the resin with or without modification using GNPs are in turn squeezed twice, spirally wound, and thermally cured to prepare the modified or unmodified C/G-HFRP bars. The operating parameters of the pultrusion/winding process have been introduced in the literature [10,19]. Finally, in this study, all C/G-HFRP bars have a nominal diameter of 8 mm and a rib spacing of 12 mm.

### 2.6. Uniaxial tensile samples

Uniaxial tensile samples of the C/G-HFRP bars were prepared as per the ASTM D7205 standard [40]. Each specimen has an overall length of 820 mm, consisting of a 40-times-diameter gauge length and two 250-mm-length anchor ends, as shown in Fig. 3(a). The specimen ends were anchored by seamless steel pipes filled with expansive agent, wherein the steel pipe had a 42 mm outer diameter and a 3 mm wall



**Fig. 3.** Uniaxial tensile tests: (a) sample dimensions; (b) loading photograph; and (c) combined strain measuring device.

thickness. In this work, the uniaxial tensile samples before and after modification using GNPs had a total of twenty-two groups, wherein each group included five identical specimens.

### 2.7. Test set-up and procedure

Uniaxial tensile tests of all samples before and after modification using GNPs were performed on a 100-ton universal testing machine. As per the ASTM D7205 standard [40], a loading rate of 0.2 kN/s was used to implement the tensile tests until each specimen was destroyed. As shown in Fig. 3(b-c), a combined strain measuring device was used to obtain the tensile strain of sample, in which two 80-mm-length strain gauges were planned to monitor the tensile strain of C/G-HFRP bars before the failure of carbon fibers, and two 30-mm-measuring-range linear variable differential transformers (LVDTs) were planned to monitor the deformation of C/G-HFRP bars after the failure of carbon fibers. Meanwhile, a DH3816 static strain acquisition system was employed to collect the tensile load and strain of all samples. The tensile strength or stress ( $\sigma$ , in MPa) of the sample can be calculated as follows:

$$\sigma = \frac{4P}{\pi D^2} \quad (3)$$

where  $P$  denotes the maximum tensile load (in N) of sample and  $D$  signifies the nominal diameter of sample (in mm).

Additionally, the physical phase analyses of GNPs and epoxy vinyl resin before and after modification were performed by XRD (PANalytical X'Pert<sup>3</sup> Power, Germany, Cu-K $\alpha$ , 40 kV and 40 mA for X-ray tube working, and a scanning speed of 0.07°/s). Simultaneously, SEM (EVO HD15, ZEISS, voltage 10 kV) was used to observe their microstructures, which were employed to explain the toughening mechanisms of GNPs-modified epoxy vinyl resin.

## 3. Results and discussion

### 3.1. XRD analysis

Fig. 4 presents the physical phase analyses of GNPs and epoxy vinyl resin before and after modification, and the inserted figure represents the analysis result of GNPs. It can be found that only one sharp diffraction peak is displayed at 26.4°, which demonstrates the composition and purity of GNPs used. As shown in Fig. 4, the epoxy vinyl resin before modification does not display any diffraction peak at 26.4°, while the one after modification shows a small diffraction peak at 26.4°. Hence, the physical phase analyses confirm the difference of

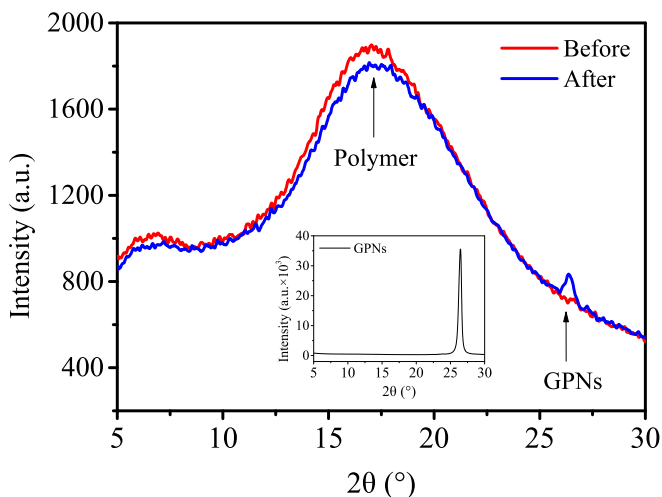


Fig. 4. Physical phase analyses of GNPs and epoxy vinyl resin before and after modification.

composition between the epoxy vinyl resin before and after modification.

### 3.2. Failure mode

The failure modes of C/G-HFRP bars under tensile load are shown in Fig. 5. Because the failure modes were identical for the samples before and after modification using GNPs, those of the modified samples are only showed here. As shown in Fig. 5, the failed samples present a global separation and breakage of the reinforced fibers within the gauge length. The fiber separation is because the energy release rate at the failure of glass fibers is larger than the fracture toughness of the epoxy vinyl resin between fibers. Besides, the sounds of fiber breakage were heard twice under an increasing tensile load; the first sound was close to  $\varepsilon_C$  and lasted for a period of time along with small and crisp sounds of fiber breakage, while the second sound was approach to  $\varepsilon_G$  and lasted for a short time along with dull and loud sounds of fiber breakage. Finally, the samples were destroyed accompanied by a loud sound under an increasing tensile load.

### 3.3. Tensile modulus

The initial tensile modulus of C/G-HFRP bar before and after modification can be predicted by the mixing rule of fibrous composites as follows [16,17,20,21]:

$$E_p = V_f(V_C E_C + V_G E_G) + E_M(1 - V_f) \quad (4)$$

where  $E_p$  is the predicted value of initial tensile modulus of sample (in GPa), and  $E_M$  is the experiment result of tensile modulus of the epoxy vinyl resin (in GPa).

The material and characteristic parameters in Tables 1 and 2 were substituted into Eq. (4) to calculate  $E_p$  of the samples before and after modification, as shown in Table 3. By comparison of  $E_p$  and  $E_E$ , adding 0.3 wt% GNPs into the epoxy vinyl resin hardly improves  $E_E$  of the modified samples. This is because the longitudinal continuous fibers primarily contribute to the tensile modulus of C/G-HFRP bars.

### 3.4. Tensile strength and strain at the initial failure of carbon fibers

In ductile C/G-HFRP bars, the failure process of carbon fibers is progressive accumulation, rising from the dispersion distribution of the tensile strength of carbon fibers, the bridging effect of surrounding glass fibers, and the dynamic redistribution of tensile stress released. Because the tensile stress-strain response of ductile C/G-HFRP bars is no longer linear elastic after the initial failure of carbon fibers, followed by plastic deformation, it is similar to the yield point of steel bar, while the working mechanisms are different from each other. In this study, the “yield strength” similar to that of steel bar is named as the tensile strength ( $\sigma_{cf}$ , in MPa) of C/G-HFRP bars at the initial failure of carbon fibers, and the corresponding strain ( $\varepsilon_{cf}$ ) is called the tensile strain at the initial failure of carbon fibers.

Herein,  $\sigma_{cf}$  of the samples before and after modification are presented in Table 4, in which  $\sigma_{cf}$  is obtained by the tensile load at the initial failure of carbon fibers dividing by the nominal cross-sectional area of sample using Eq. (3). As shown in Table 4,  $\sigma_{cf}$  of almost all samples are around 900 MPa, which belongs to a high-strength category relative to steel bar. This stems  $E_E$  of C/G-HFRP bars from that is approximately one third of that of steel bar, but  $\varepsilon_{cf}$  is about six times that of steel bar.

Fig. 6(a)–(c) present the comparison of  $\sigma_{cf}$  between the samples before and after modification using GNPs. Firstly,  $\sigma_{cf}$  of all samples increased with  $V_C:V_G$ . This is because an increasing  $V_C:V_G$  improves  $E_E$  of the sample. Similarly,  $\sigma_{cf}$  of all samples also increased with  $V_f$ , resulting from an increasing  $E_E$ , as shown in Fig. 6(c).

Obviously, the addition of GNPs improved  $\sigma_{cf}$  of the C/G-HFRP bars. As shown in Fig. 6(a) and (b), for the modified samples with a

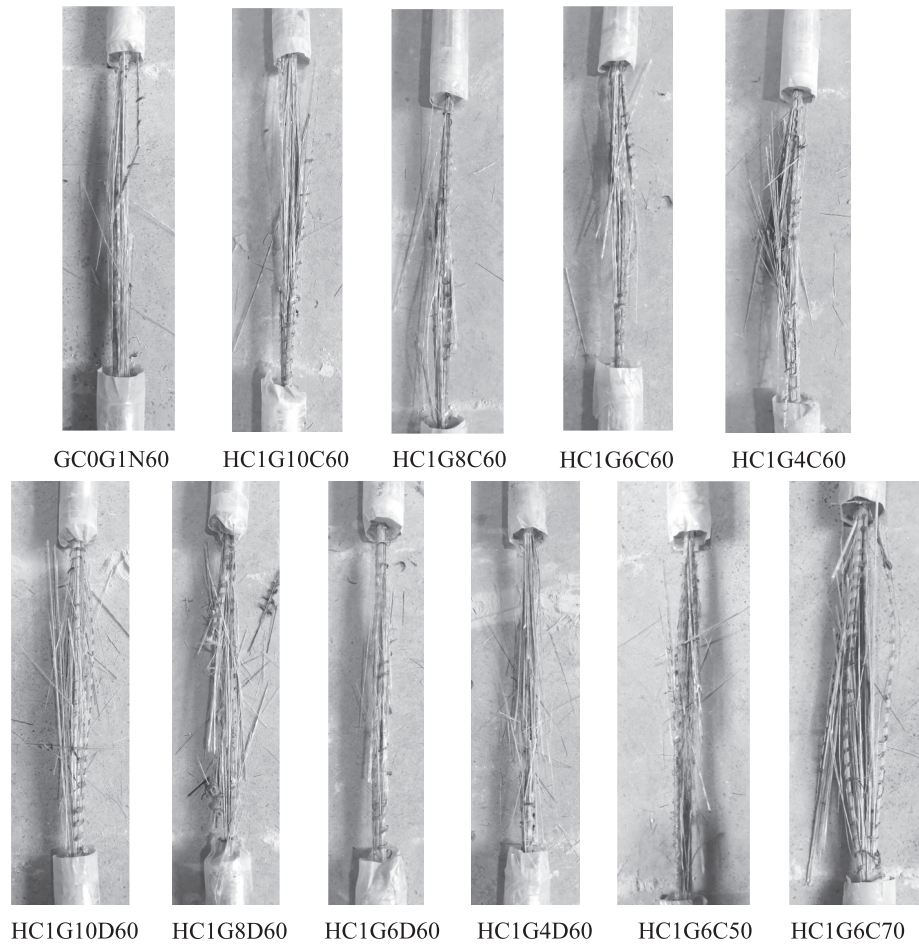


Fig. 5. Failure modes of the samples after modification under tensile load.

Table 3

Comparison of tensile moduli between the samples before and after modification.

| Sample code | Experiment results (GPa) |               | Predicted values (GPa) | $E_{E-After}/E_{E-Before}$ | $E_{E-After}/E_P$ |
|-------------|--------------------------|---------------|------------------------|----------------------------|-------------------|
|             | $E_{E-Before}$           | $E_{E-After}$ | $E_P$                  |                            |                   |
| GC0G1N60    | 50.6 ± 0.83              | 50.7 ± 1.96   | 51.0                   | 1.002                      | 0.994             |
| HC1G10C60   | 66.8 ± 0.60              | 67.5 ± 2.47   | 67.0                   | 1.011                      | 1.007             |
| HC1G8C60    | 69.5 ± 0.87              | 70.8 ± 1.43   | 69.9                   | 1.019                      | 1.013             |
| HC1G6C60    | 73.7 ± 1.41              | 73.8 ± 1.59   | 72.7                   | 1.001                      | 1.015             |
| HC1G4C60    | 84.5 ± 0.99              | 84.8 ± 1.73   | 86.0                   | 1.004                      | 0.986             |
| HC1G10D60   | 67.2 ± 0.75              | 66.2 ± 1.47   | 67.0                   | 0.985                      | 0.988             |
| HC1G8D60    | 69.5 ± 0.70              | 69.8 ± 1.72   | 69.9                   | 1.004                      | 0.999             |
| HC1G6D60    | 72.6 ± 0.22              | 73.0 ± 1.66   | 72.7                   | 1.006                      | 1.004             |
| HC1G4D60    | 85.2 ± 0.79              | 86.5 ± 2.34   | 86.0                   | 1.015                      | 1.006             |
| HC1G6C50    | 63.4 ± 0.70              | 63.5 ± 1.99   | 63.4                   | 1.002                      | 1.002             |
| HC1G6C70    | 85.4 ± 0.90              | 84.9 ± 2.39   | 86.0                   | 0.994                      | 0.987             |
| Average     |                          |               |                        | 1.004                      | 1.000             |

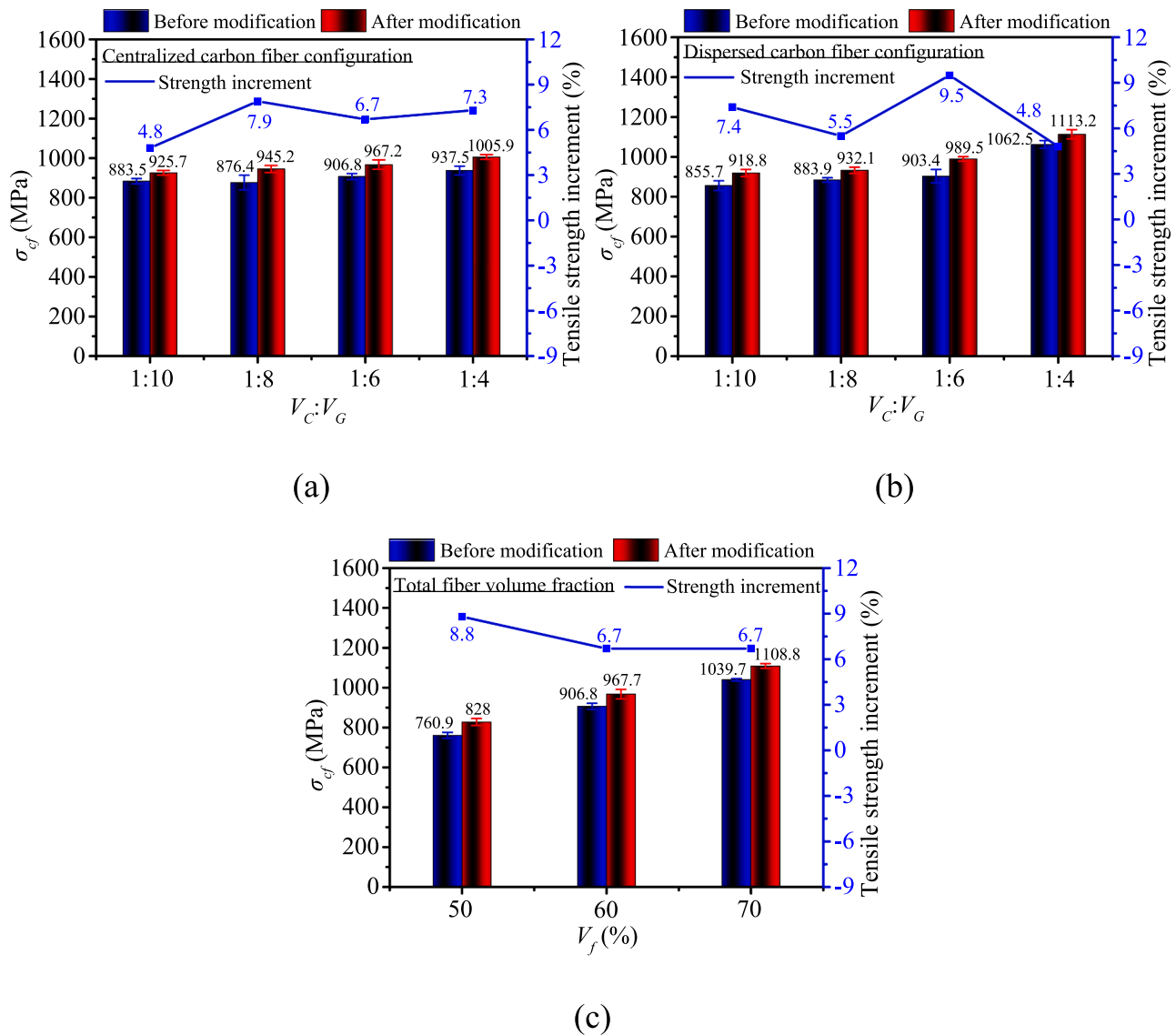
Note:  $E_{E-Before}$  and  $E_{E-After}$  are the experiment results of initial tensile moduli of the samples before and after modification, respectively.

Table 4

Tensile characteristics of the samples at the initial failure of carbon fibers.

| Sample code | $\sigma_{cf}$ (MPa)  |                     |                                        | $\epsilon_{cf}$ (%)    |                       |                                            |
|-------------|----------------------|---------------------|----------------------------------------|------------------------|-----------------------|--------------------------------------------|
|             | $\sigma_{cf-Before}$ | $\sigma_{cf-After}$ | $\sigma_{cf-After}/\sigma_{cf-Before}$ | $\epsilon_{cf-Before}$ | $\epsilon_{cf-After}$ | $\epsilon_{cf-After}/\epsilon_{cf-Before}$ |
| HC1G10C60   | 883.5 ± 13.6         | 925.7 ± 11.9        | 1.048                                  | 1.333 ± 0.04           | 1.368 ± 0.03          | 1.026                                      |
| HC1G8C60    | 876.4 ± 36.9         | 945.2 ± 18.0        | 1.079                                  | 1.286 ± 0.12           | 1.337 ± 0.01          | 1.039                                      |
| HC1G6C60    | 906.8 ± 15.1         | 967.7 ± 23.9        | 1.067                                  | 1.214 ± 0.01           | 1.289 ± 0.08          | 1.061                                      |
| HC1G4C60    | 937.5 ± 22.2         | 1005.9 ± 12.6       | 1.073                                  | 1.158 ± 0.05           | 1.191 ± 0.07          | 1.028                                      |
| HC1G10D60   | 855.7 ± 24.1         | 918.8 ± 18.1        | 1.074                                  | 1.279 ± 0.05           | 1.356 ± 0.02          | 1.060                                      |
| HC1G8D60    | 883.9 ± 11.0         | 932.1 ± 16.0        | 1.055                                  | 1.248 ± 0.01           | 1.342 ± 0.01          | 1.075                                      |
| HC1G6D60    | 903.4 ± 33.4         | 989.5 ± 11.5        | 1.095                                  | 1.291 ± 0.09           | 1.335 ± 0.06          | 1.034                                      |
| HC1G4D60    | 1062.5 ± 19.4        | 1113.2 ± 22.9       | 1.048                                  | 1.264 ± 0.05           | 1.276 ± 0.03          | 1.010                                      |
| HC1G6C50    | 760.9 ± 14.9         | 828.0 ± 17.3        | 1.088                                  | 1.221 ± 0.06           | 1.288 ± 0.06          | 1.054                                      |
| HC1G6C70    | 1039.7 ± 6.3         | 1108.8 ± 11.8       | 1.067                                  | 1.244 ± 0.03           | 1.303 ± 0.09          | 1.047                                      |
| Average     |                      |                     | 1.069                                  |                        |                       | 1.043                                      |

Note:  $\sigma_{cf-Before}$  and  $\sigma_{cf-After}$  are the experiment results of tensile strengths of the samples before and after modification, respectively.  $\epsilon_{cf-Before}$  and  $\epsilon_{cf-After}$  are the corresponding tensile strains, respectively.

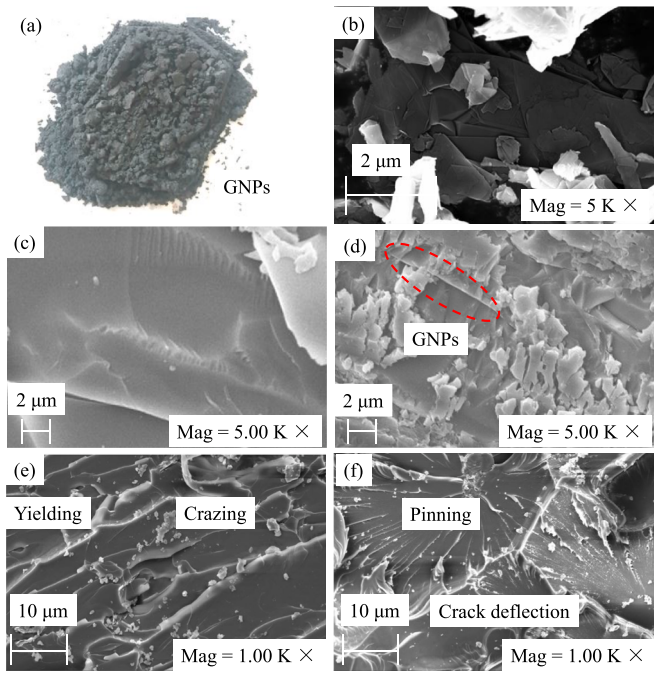


**Fig. 6.** Comparison of  $\sigma_{cf}$  between the samples before and after modification: (a) centralized carbon fiber configuration; (b) dispersed carbon fiber configuration; and (c) total fiber volume fraction.

centralized carbon fiber configuration,  $\sigma_{cf}$  were raised by 4.8%, 7.9%, 6.7% and 7.3% with an increasing  $V_C:V_G$  from 1:10 to 1:4 relative to those of the samples without modification using GNPs, while  $\sigma_{cf}$  were heightened by 7.4%, 5.5%, 9.5% and 4.8% for the modified samples with a dispersed carbon fiber configuration, respectively. As shown in Fig. 6(c), for the modified samples with  $V_f$  of 50%, 60% and 70%,  $\sigma_{cf}$  were improved by 8.8%, 6.7% and 6.7%, respectively. In general,  $\sigma_{cf}$  of all samples were averagely improved by 6.9% after modification using GNPs. Comparatively, the increment of  $\sigma_{cf}$  for the modified samples with a centralized carbon fiber configuration was averagely 0.125% lower than that of the counterparts with a dispersed carbon fiber configuration. The improvement of  $\sigma_{cf}$  stems incorporating 0.3 wt% GNPs into the polymer matrix from that obtains an improvement of 13.0% and 22.1% for the tensile modulus and fracture toughness of the modified polymer matrix, respectively.

Fig. 7(a) and (b) present the stacked GNPs powder and their microstructures by SEM, respectively. It is observed that there are wrinkles on the surface of GNPs, realizing a self-locking function with the polymer matrix. Apparently, the fracture morphology of plain epoxy vinyl resin (without modification using GNPs) was smooth and featureless due to brittle failure (Fig. 7(c)), while that of the modified one appeared rough

due to crack deflection (Fig. 7(d)–(f)). These results are evidences of toughness improvement for the GNPs-modified epoxy vinyl resin. It was reported that the crack deflection was a primary toughening mechanism for the large aspect-ratio GNPs-modified polymer matrix [41]. Normally, if crack tip encounters rigid GNPs oriented randomly in polymer matrix, the crack generates deflection to form disturbed cracking routes, which magnify the energy consumption and the fracture toughness of the modified polymer resin. Moreover, the crack deflection typically originates from three situations when the crack tip in the matrix resin encounters rigid GNPs oriented randomly [41,42]. As shown in Fig. 8(a), when encountering the GNPs oriented at an angle, the crack route propagates along the GNP surface to cause crack deflection and layer breakage. As shown in Fig. 8(b), if the crack tip is parallel to the GNPs oriented, the flakes are forced to separate in-between the GNPs due to a low Van der Waals force. Additionally, when the crack tip encounters perpendicularly the GNPs oriented, the crack tip is deflected by the GNP and goes around it where meets the flakes edges, and then the crack tip could de-laminate the flakes to create a “V” shaped GNPs with layer breakage, as shown in Fig. 8(c). Also, the yielding, crazing, pinning of the modified polymer matrix (Fig. 7(e) and (f)), and the debonding of GNPs simultaneously contributed to the energy consumption capacity



**Fig. 7.** Microstructures of GNPs and section morphologies after failure (a) stacked GNPs powder; (b) magnified GNPs; (c) plain polymer matrix; (d) GNPs oriented randomly; (e) yielding and crazing; and (f) pinning and crack deflection.

and cracking resistance, which delayed the time of failure occurrence. Therefore, it is credible that addition of 0.3 wt% GNPs significantly dissipates more energy and improves the fracture toughness of epoxy vinyl resin.

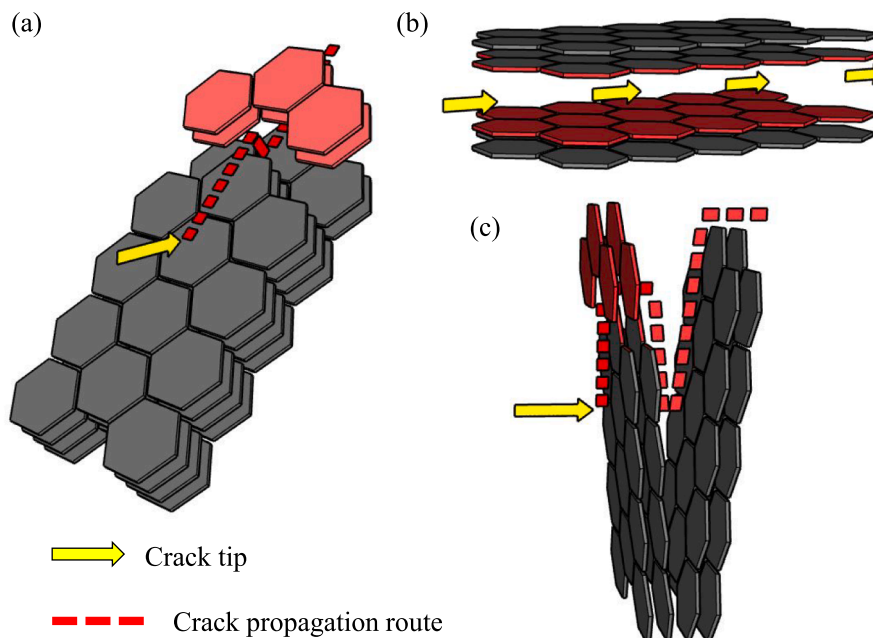
Fig. 9(a)–(c) show the comparison of  $\varepsilon_{cf}$  between the samples before and after modification. Obviously,  $\varepsilon_{cf}$  of the modified samples was larger than that of the unmodified ones. For the modified samples with a centralized carbon fiber configuration,  $\varepsilon_{cf}$  were improved by 4.0%, 4.0%, 6.2% and 2.9% with an increasing  $V_C:V_G$  from 1:10 to 1:4 relative to those of the unmodified ones, respectively, while  $\varepsilon_{cf}$  were raised by

6.0%, 7.5%, 3.4% and 1.0% for the modified samples with a dispersed carbon fiber configuration. Meanwhile,  $\varepsilon_{cf}$  were heightened by 5.5%, 6.2% and 4.7% for the modified samples with  $V_f$  of 50%, 60% and 70%, respectively. In general,  $\varepsilon_{cf}$  of the samples after modification using GNPs were averagely improved by 4.7%, wherein the increment of  $\varepsilon_{cf}$  for the modified samples with a centralized carbon fiber configuration was averagely 0.2% lower than that of the counterparts with a dispersed carbon fiber configuration.

### 3.5. Ultimate tensile strength and strain

The maximum tensile strength that the sample can undertake until failure is called the ultimate tensile strength ( $\sigma_u$ , in MPa), while the corresponding strain is called the ultimate tensile strain ( $\varepsilon_u$ ).  $\sigma_u$  of the samples before and after modification using GNPs are presented in Table 5, where  $\sigma_u$  was determined by the maximum tensile load dividing by the nominal cross-sectional area of sample using Eq. (3). As shown in Fig. 10(a) and (b),  $\sigma_u$  of the sample decreases with an increasing  $V_C:V_G$  from 0:1 to 1:4. Simultaneously,  $\sigma_u$  of the sample goes up with an increasing  $V_f$  from 50% to 70%, as shown in Fig. 10(c). This is because the glass fibers finally fail in the samples under tensile load due to a high ultimate strain, so  $\sigma_u$  of the sample is associated with  $V_G$ .

Additionally, as shown in Fig. 10(a)–(c), for the modified samples with a centralized carbon fiber configuration,  $\sigma_u$  were improved by 4.6%, 3.0%, 3.8%, 7.7% and 14.9% with an increasing  $V_C:V_G$  from 0:1 to 1:4 relative to that of the unmodified ones, respectively, while  $\sigma_u$  were improved by 4.6%, 6.3%, 8.8%, 2.9% and 3.4% for the modified samples with a dispersed carbon fiber configuration. Meanwhile,  $\sigma_u$  were increased by 7.3%, 7.7% and 15.7% for the modified samples with  $V_f$  of 50%, 60% and 70%, respectively. The improvement of  $\sigma_u$  stems incorporating 0.3 wt% GNPs into the polymer matrix from that enhances the fracture toughness of the resin between fibers. The polymer matrix toughened can withstand the energy release rate of C/G-HFRP bar at the failure of carbon fibers to reduce the separation degree of glass fibers around the failed carbon fibers. Consequently, the global structure of modified samples after the complete failure of carbon fibers was more intact than that of the unmodified samples. The intact glass fiber structure represents a good cooperative working mechanism, which delays the ultimate state of modified samples under tensile load.



**Fig. 8.** Schematic diagram of interaction between crack tip and GNPs: (a) at an angle; (b) parallel; and (c) perpendicular.

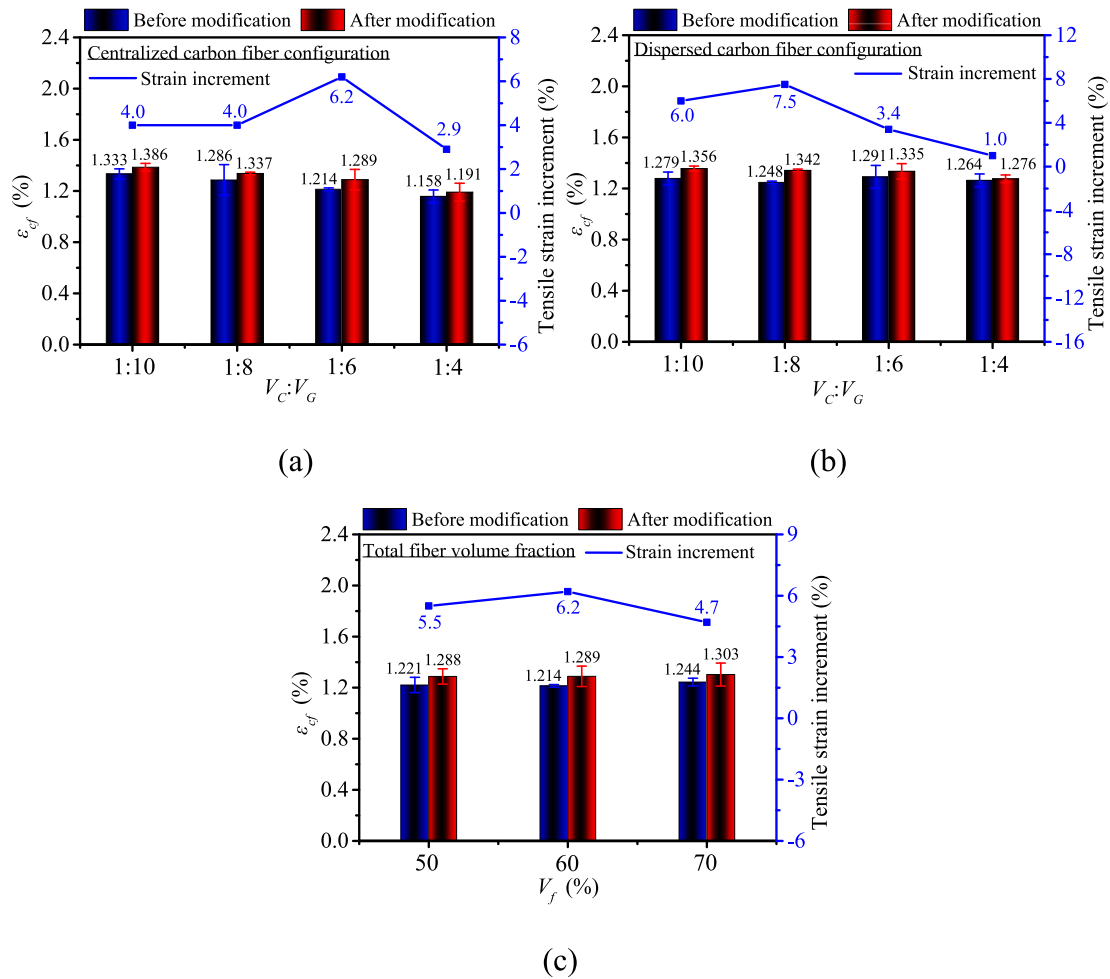


Fig. 9. Comparison of  $\epsilon_{cf}$  between the samples before and after modification: (a) centralized carbon fiber configuration; (b) dispersed carbon fiber configuration; and (c) total fiber volume fraction.

**Table 5**  
Ultimate tensile characteristics of the samples before and after modification.

| Sample code | $\sigma_u$ (MPa)    |                    |                                      | $\epsilon_u$ (%)      |                      |                                          |
|-------------|---------------------|--------------------|--------------------------------------|-----------------------|----------------------|------------------------------------------|
|             | $\sigma_{u-Before}$ | $\sigma_{u-After}$ | $\sigma_{u-After}/\sigma_{u-Before}$ | $\epsilon_{u-Before}$ | $\epsilon_{u-After}$ | $\epsilon_{u-After}/\epsilon_{u-Before}$ |
| GC0G1N60    | 1297.6<br>± 29.7    | 1357.1<br>± 65.1   | 1.046                                | 2.564<br>± 0.05       | 2.677<br>± 0.12      | 1.044                                    |
| HC1G10C60   | 1199.0<br>± 45.0    | 1234.9<br>± 63.5   | 1.030                                | 2.635<br>± 0.09       | 2.714<br>± 0.14      | 1.030                                    |
| HC1G8C60    | 1111.5<br>± 72.2    | 1154.2<br>± 55.5   | 1.038                                | 2.628<br>± 0.17       | 2.729<br>± 0.13      | 1.038                                    |
| HC1G6C60    | 1029.1<br>± 32.2    | 1108.0<br>± 18.7   | 1.076                                | 2.382<br>± 0.07       | 2.565<br>± 0.04      | 1.076                                    |
| HC1G4C60    | 949.0 ± 30.5        | 1090.2 ± 40.3      | 1.149                                | 2.270<br>± 0.07       | 2.608<br>± 0.09      | 1.149                                    |
| HC1G10D60   | 1170.6<br>± 25.0    | 1244.5<br>± 28.4   | 1.063                                | 2.573<br>± 0.05       | 2.735<br>± 0.06      | 1.063                                    |
| HC1G8D60    | 1105.1<br>± 26.1    | 1202.4<br>± 40.7   | 1.088                                | 2.613<br>± 0.06       | 2.842<br>± 0.09      | 1.088                                    |
| HC1G6D60    | 1085.5<br>± 52.9    | 1116.5<br>± 73.4   | 1.028                                | 2.513<br>± 0.12       | 2.585<br>± 0.16      | 1.028                                    |
| HC1G4D60    | 1062.5<br>± 19.4    | 1098.5<br>± 42.7   | 1.034                                | 2.542<br>± 0.04       | 2.628<br>± 0.10      | 1.034                                    |
| HC1G6C50    | 904.4 ± 25.5        | 970.2 ± 37.3       | 1.072                                | 2.498<br>± 0.07       | 2.680<br>± 0.10      | 1.072                                    |
| HC1G6C70    | 1094.1<br>± 64.4    | 1262.0<br>± 39.0   | 1.153                                | 2.373<br>± 0.13       | 2.737<br>± 0.08      | 1.153                                    |
| Average     |                     |                    | 1.071                                |                       |                      | 1.071                                    |

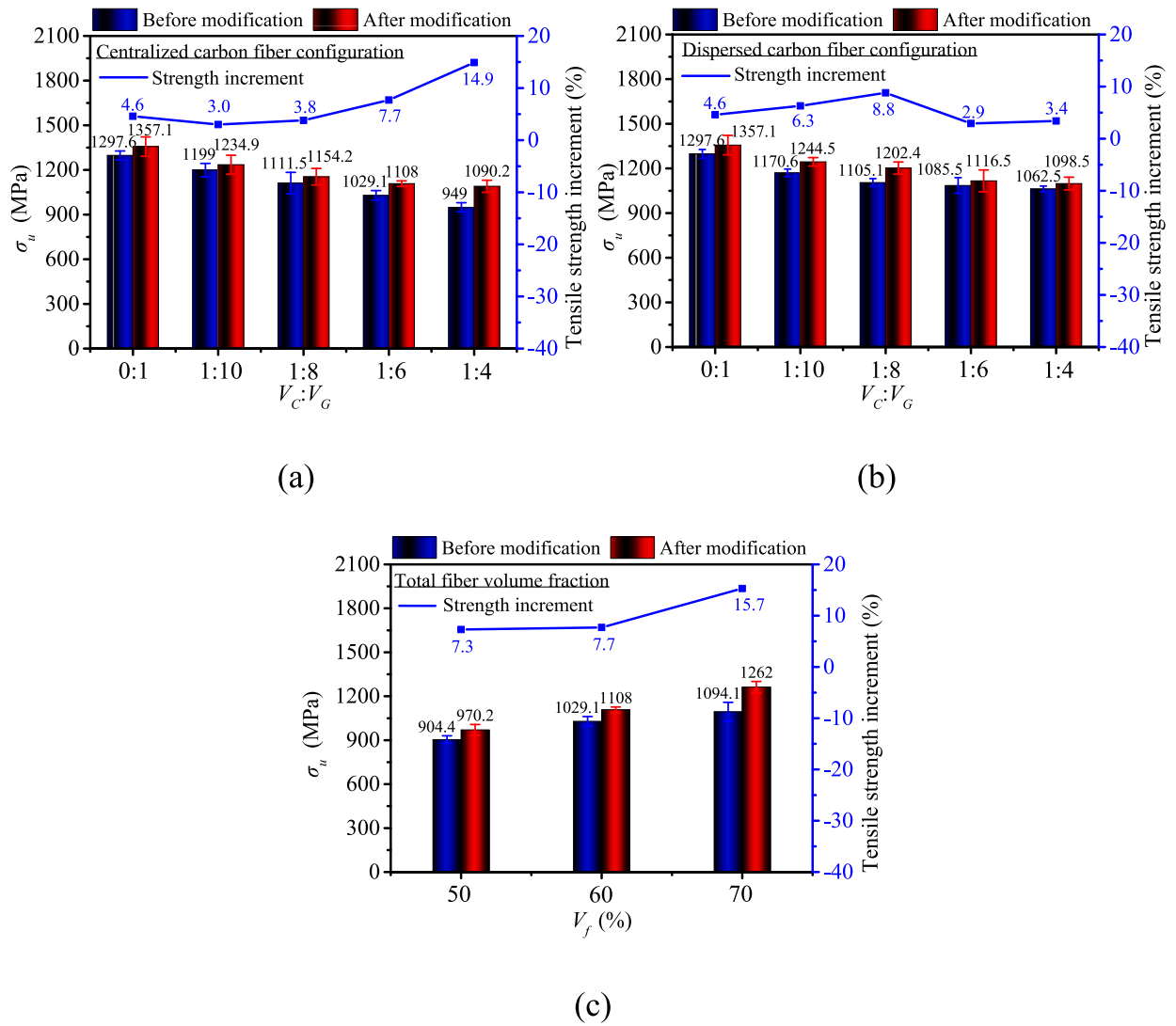
However, the average increment of  $\sigma_u$  for the modified samples with a centralized carbon fiber configuration was 1.6% higher than that of the counterparts with a dispersed carbon fiber configuration. This is because the dispersed carbon fibers after failure destroy the global structure of glass fibers to some extent, affecting their collaboration ability under an increasing tensile load.

Note that the peripheral glass fibers in the modified samples retain their global structure after the complete failure of carbon fibers; hence, the back-to-back strain gauges pasted on glass fibers were not damaged until the modified samples reached their ultimate tensile state. In contrast, the strain gauges pasted on the unmodified samples were completely destroyed, because the global structure of the peripheral glass fibers presents a severe separation after the failure of carbon fibers. Herein, the combined LVDT data was used to depict approximately the tensile stress–strain responses of the unmodified samples. However, collected  $\epsilon_u$  did not accurately represent the real result of testing samples. Depending on  $\sigma_u$  of C/G-HFRP bars and the remaining tensile modulus ( $E_{sec}$ , in GPa, called secondary tensile stiffness) after the complete failure of carbon fibers,  $\epsilon_u$  can be determined as follows:

$$\epsilon_u = \frac{\sigma_u}{E_{sec}} = \frac{\sigma_u}{V_f V_G E_G + E_M (1 - V_f)} \quad (5)$$

where  $E_{sec}$  is equal to the slope of straight line at the strain hardening stage in the tensile stress–strain response of C/G-HFRP bar.

As shown in Table 5,  $\epsilon_u$  determined using Eq. (5) is in accordance with  $\epsilon_G$  (2.5%) of glass fibers provided by the manufacturer. Fig. 11(a)–(c) present the comparison of  $\epsilon_u$  for the samples before and after



**Fig. 10.** Comparison of  $\sigma_u$  between the samples before and after modification: (a) centralized carbon fiber configuration; (b) dispersed carbon fiber configuration; and (c) total fiber volume fraction.

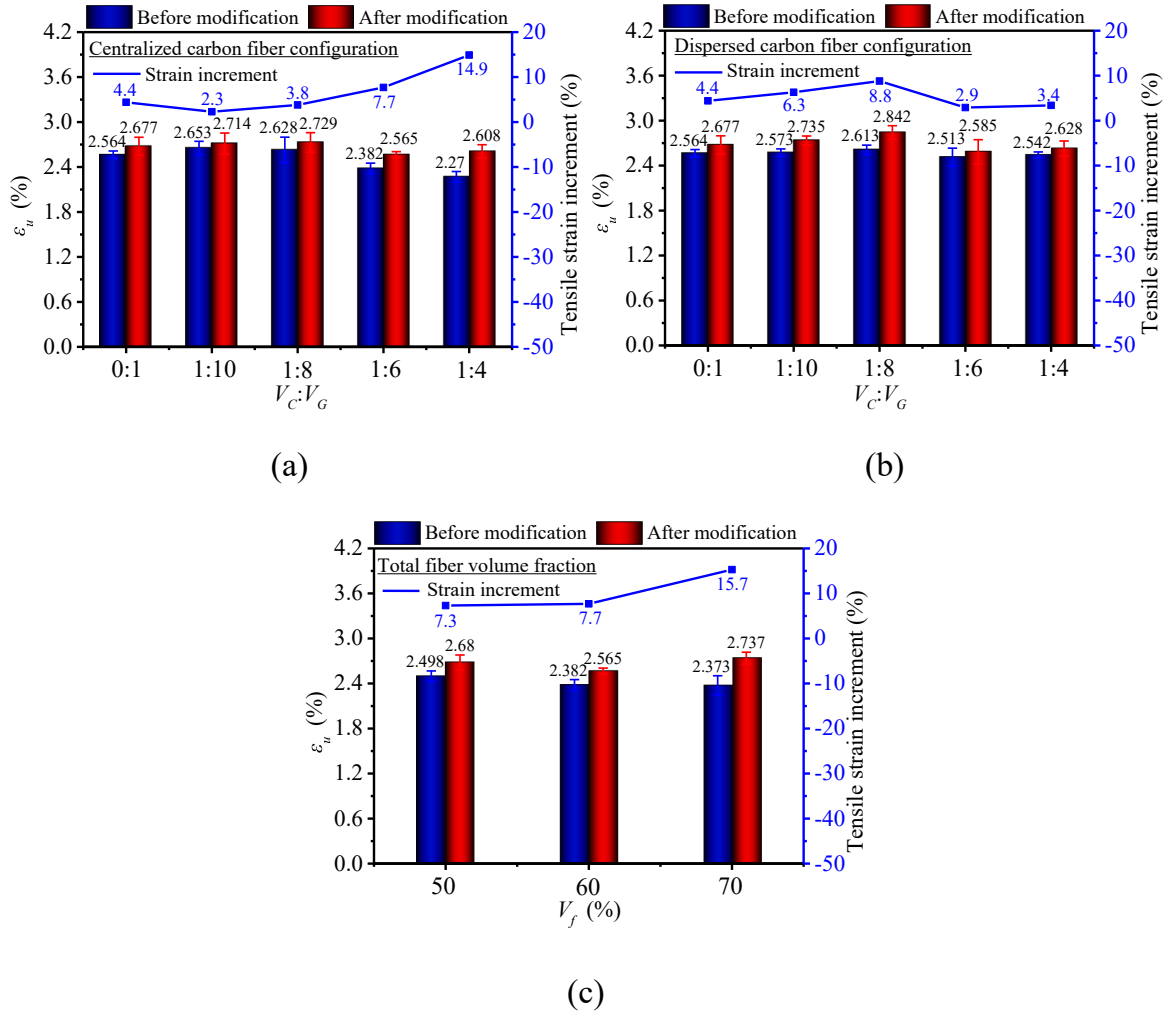
modification using GNPs. It is clear that  $\epsilon_u$  of the modified samples is larger than that of the unmodified ones, and this also verifies the aforementioned motivation analysis. In fact, the contribution of the tensile modulus of C/G-HFRP bars primarily originates from longitudinal continuous fibers irrespective of the modified polymer matrix, as shown in Table 3; in other word, the addition of GNPs into the polymer matrix has almost no impact on the tensile modulus of modified C/G-HFRP bars. Therefore, because of an almost identical  $E_{sec}$ ,  $\epsilon_u$  of the modified samples had a similar increment to that of  $\sigma_u$  relative to the unmodified samples.

### 3.6. Tensile stress–strain responses

For the C/G-HFRP bars before modification using GNPs, the tensile strain was presented by combining the values measured by strain gauges in linear elastic stage with those measured by LVDTs in plastic deformation stage. In this way, although the strain gauges are prematurely destroyed by the strain energy released of C/G-HFRP bar at the failure of carbon fibers under tensile load, the tensile stress–strain responses in plastic deformation stage can be ensured by LVDTs.

Using the aforementioned combination method, the typical tensile stress–strain responses of the samples before modification are exhibited in Fig. 12(a)–(d). The tensile stress of the sample GC0G1N60 linearly

increases with the tensile strain until it is destroyed under tensile load (Fig. 12(a)), so its failure mode is brittle. Comparatively, the tensile stress–strain responses of the samples HC1G4C60 and HC1G4D60 consist of a linear elastic stage and a jagged plateau (Fig. 12(b) and (c)). In the linear elastic stage, the tensile stress linearly increases with the tensile strain until the carbon fibers begin to fail. Subsequently, a jagged plateau is presented in the tensile stress–strain responses, resulting from that the carbon fibers progressively break under an increasing tensile load. Moreover,  $V_C:V_G$  (1:4.34) in the samples HC1G4C60 and HC1G4D60 is close to the critical ratio (1:4.30), indicating that the glass fibers no longer have additional load-bearing capacity to undertake an increasing tensile load after completely taking load transferred from the failed carbon fibers. Consequently, a strain hardening stage does not appear in their tensile stress–strain responses. Similarly, for the samples with  $V_C:V_G$  ranging from 1:10 to 1:6, the tensile stress–strain responses also contain a linear elastic stage and a jagged plateau, in which the variation between the tensile stress and strain has a similar trend to those of the samples HC1G4C60 and HC1G4D60 (Fig. 12(b) and (c)). However, because  $V_C:V_G$  is lower than 1:4.30 for these samples, their tensile stress–strain responses also exhibit a strain hardening stage, in which the tensile stress linearly increases with the tensile strain (Fig. 12(b)–(d)). Therefore, the failure modes of the samples with  $V_C:V_G$  ranging from 1:10 to 1:4 are ductile.



**Fig. 11.** Comparison of  $\epsilon_u$  between the samples before and after modification: (a) centralized carbon fiber configuration; (b) dispersed carbon fiber configuration; and (c) total fiber volume fraction.

Using only strain gauge data, the tensile stress–strain responses of the C/G-HFRP bars after modification using GNPs are shown in Figs. 13(a)–(d). The strain gauges are not destroyed until the ultimate state of glass fibers is reached, so a relative complete tensile stress–strain response can be presented in terms of the strain gauge data. The integrity of strain gauges stems a global structure of the peripheral glass fibers from that protects them from damaging after the failure of carbon fibers. The tensile stress–strain responses of the samples after modification have a similar development trend to those of the ones before modification using GNPs (Figs. 13(a)–(d)). Nevertheless, for the tensile stress–strain responses of the samples after modification, the stress drop rate at the initial failure of carbon fibers was significantly reduced and the fluctuation of jagged plateau tended to be flatter.

#### 4. Evaluation indicators

##### 4.1. Hybrid effect

In this study,  $\epsilon_C$  of the carbon fibers is 1.2%, which is provided by the manufacturer. Studies indicated that the ultimate tensile strain ( $\epsilon_{cbar}$ ) of the carbon-FRP bar was 15 ~ 20% lower than  $\epsilon_C$  of the corresponding carbon fibers, resulting from the size effect of fibrous composite [16,21,43]. Because the number of carbon fibers was limited in the laboratory, the corresponding carbon-FRP bar was not prepared in this work. Based on the previous research conclusions,  $\epsilon_{cbar}$  is considered to

be 0.984%, which is 18% lower than  $\epsilon_C$  of the corresponding carbon fibers. However, due to the hybrid effect of fibrous composite, the failure strain of the carbon fibers in C/G-HFRP bar was further improved. The hybrid effect (HE) can be quantified as follows:

$$HE = \frac{\epsilon_{cf} - \epsilon_{cbar}}{\epsilon_{cbar}} \quad (6)$$

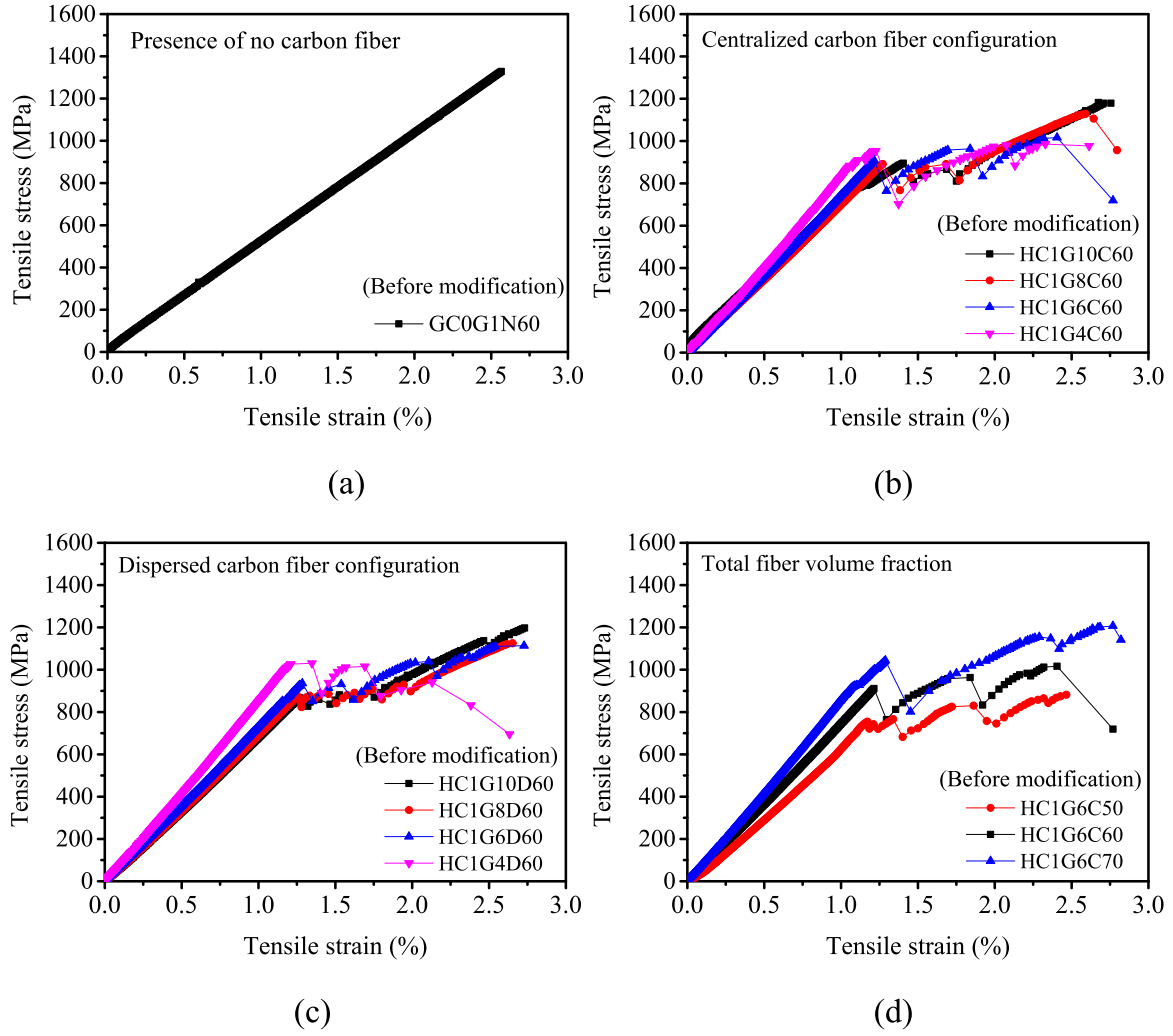
Table 6 presents the quantitative results of HE for the samples before and after modification using GNPs. An average increment of 20.9% was obtained for the samples after modification relative to the ones before modification.

##### 4.2. Ductile strain

Ductile strain means that the difference between  $\epsilon_u$  and another tensile strain ( $\sigma_u/E_E$ ), and the latter is determined by prolonging the initial straight line in the stress–strain response and intersecting at  $\sigma_u$  level (Fig. 14). The ductile strain ( $\epsilon_{DS}$ ) can be quantified as follows:

$$\epsilon_{DS} = \frac{\sigma_u}{E_E} - \epsilon_u \quad (7)$$

Table 6 presents the quantitative results of  $\epsilon_{DS}$  for the samples before and after modification using GNPs. An average increment of 7.9% was obtained for the samples after modification relative to the ones before modification.



**Fig. 12.** Typical tensile stress–strain responses of the samples before modification: (a) presence of no carbon fiber; (b) centralized carbon fiber configuration; (c) dispersed carbon fiber configuration; and (d) total fiber volume fraction.

#### 4.3. Stress drop rate

A stress drop is shown in the stress–strain responses of C/G-HFRP bars at the initial failure of carbon fibers (Fig. 14). The presence of stress drop rises from stress dynamic redistribution, which comes from the progressive damage of carbon fibers under an increasing tensile load. The stress drop is unfavorable for C/G-HFRP bar-reinforced concrete members. Based on the experimental results, the stress drop ( $R_S$ ) can be quantified as follows:

$$R_S = \frac{\sigma_{cf} - \sigma_{p-cf}}{\sigma_{cf}} \quad (8)$$

where  $\sigma_{p-cf}$  is the tensile strength after stress drop at the initial failure of carbon fibers.

Table 6 presents the quantitative results of  $R_S$  for the samples before and after modification using GNPs. An average reduction of 52.4% was obtained for the samples after modification relative to the ones before modification.

#### 4.4. Strength ratio

For traditional steel bars, the ratio of the ultimate strength to yield strength is normally employed to evaluate their ability of plastic deformability under tensile load. It is an important indicator to measure the safety margins of reinforced concrete members. Herein, the ratio

( $R_{u/cf}$ ) of  $\sigma_u$  to  $\sigma_{cf}$  is used to measure the ability of plastic deformability of C/G-HFRP bars under tensile load.  $R_{u/cf}$  can be quantified as follows:

$$R_{u/cf} = \frac{\sigma_u}{\sigma_{cf}} \quad (9)$$

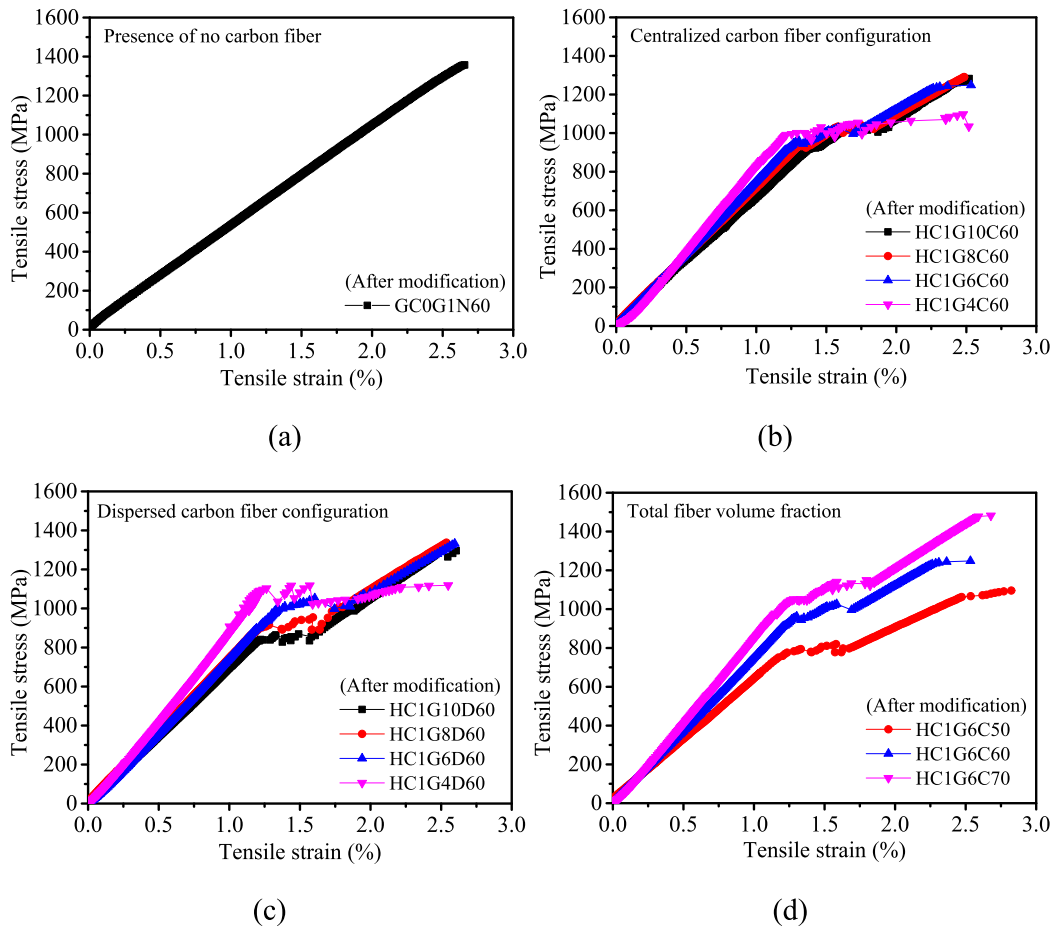
Table 6 presents the quantitative results of  $R_{u/cf}$  for the samples before and after modification using GNPs. An average increment of 0.4% was obtained for the samples after modification relative to the ones before modification.

In general, the addition of GNPs improved the fracture toughness of polymer matrix, which increased the ability of energy consumption and crack resistance for the modified polymer matrix under applied load. As a result, the ability of cooperative work between the carbon and glass fibers was strengthened.

### 5. Secondary stiffness analysis

The secondary stiffness ( $E_{sec}$ ) refers to the remaining tensile modulus of C/G-HFRP bar after the complete failure of carbon fibers, which is equal to the slope of straight line at the strain hardening stage in the tensile stress–strain response (Fig. 15). For the concrete beams with plastic hinge, the secondary stiffness of C/G-HFRP bars plays a crucial role.

Xu et al. [44–46] surmised that the failed high-stiffness fibers with a critical length in HFRP bar continued to undertake part of tensile load.



**Fig. 13.** Typical tensile stress–strain responses of the samples after modification: (a) presence of no carbon fiber; (b) centralized carbon fiber configuration; (c) dispersed carbon fiber configuration; and (d) total fiber volume fraction.

**Table 6**

Quantitative results of evaluation indicators.

| Sample code | $HE$   |       |           | $\varepsilon_{DS}$ |       |           | $R_S$  |       |           | $R_{u/cf}$ |       |           |
|-------------|--------|-------|-----------|--------------------|-------|-----------|--------|-------|-----------|------------|-------|-----------|
|             | Before | After | Increment | Before             | After | Increment | Before | After | Reduction | Before     | After | Increment |
| HC1G10C60   | 35.5   | 39.0  | 9.9       | 0.84               | 0.89  | 6.0       | 10.9   | 3.4   | 68.8      | 1.357      | 1.334 | −1.7      |
| HC1G8C60    | 30.7   | 35.9  | 16.9      | 1.03               | 1.10  | 6.8       | 11.9   | 5.1   | 57.1      | 1.269      | 1.221 | −3.8      |
| HC1G6C60    | 23.4   | 31.0  | 32.5      | 0.99               | 1.06  | 7.1       | 19.1   | 6.1   | 68.1      | 1.135      | 1.145 | 0.9       |
| HC1G4C60    | 17.7   | 21.0  | 18.6      | 1.15               | 1.32  | 14.8      | 25.3   | 8.1   | 68.0      | 1.013      | 1.084 | 7.0       |
| HC1G10D60   | 30.0   | 37.8  | 26.0      | 0.83               | 0.86  | 3.6       | 4.2    | 3.8   | 9.5       | 1.368      | 1.355 | −1.0      |
| HC1G8D60    | 26.8   | 36.4  | 35.8      | 1.02               | 1.12  | 9.8       | 6.8    | 6.0   | 11.8      | 1.250      | 1.290 | 3.2       |
| HC1G6D60    | 31.2   | 35.7  | 14.4      | 1.02               | 1.06  | 3.9       | 10.2   | 6.3   | 38.2      | 1.202      | 1.128 | −6.2      |
| HC1G4D60    | 28.5   | 29.7  | 4.2       | 1.30               | 1.36  | 4.6       | 16.7   | 7.3   | 56.3      | 1.000      | 0.987 | −1.3      |
| HC1G6C50    | 24.1   | 30.9  | 28.2      | 1.07               | 1.15  | 7.5       | 14.3   | 4.3   | 69.9      | 1.189      | 1.172 | −1.4      |
| HC1G6C70    | 26.4   | 32.4  | 22.7      | 1.09               | 1.25  | 14.7      | 20.8   | 4.9   | 76.4      | 1.053      | 1.138 | 8.1       |
| Average     |        |       | 20.9      |                    |       | 7.9       |        |       | 52.4      |            |       | 0.4       |

This is because the high-stiffness fibers with a low tensile strain firstly fail under tensile load. Therefore, Xu et al. suggested that the secondary stiffness of HFRP bar could be determined as follows:

$$E_{sec} = V_f \left( \frac{V_H E_H}{2} + V_L E_L \right) + E_M (1 - V_f) \quad (10)$$

where  $E_L$  and  $E_H$  are the tensile moduli of low-stiffness and high-stiffness fibers in HFRP bar (in GPa), respectively;  $V_L$  and  $V_H$  are the volume fractions of low-stiffness and high-stiffness fibers in HFRP bar, respectively.

It can be seen from Eq. (10) that half of the failed high-stiffness fibers in HFRP bar continue to contribute to the tensile modulus, while Swolfs

et al. reported that an approximate 30% of the failed clusters were observed to be non-coplanar in the damaged fibrous composite by synchrotron radiation computed tomography [47]. It seems to indicate that 30% of the failed high-stiffness fibers in HFRP bar still participate in the final work. Additionally, some researchers ignored completely the contribution of failed high-stiffness fibers to the tensile constitutive law at the strain hardening stage in the stress–strain response of HFRP bar [17,21]. In general, the secondary stiffness of HFRP bar can be expressed as follows:

$$E_{sec} = V_f (\alpha V_H E_H + V_L E_L) + E_M (1 - V_f) \quad (11)$$

where  $\alpha$  is a contribution coefficient of high-stiffness fibers after failure.

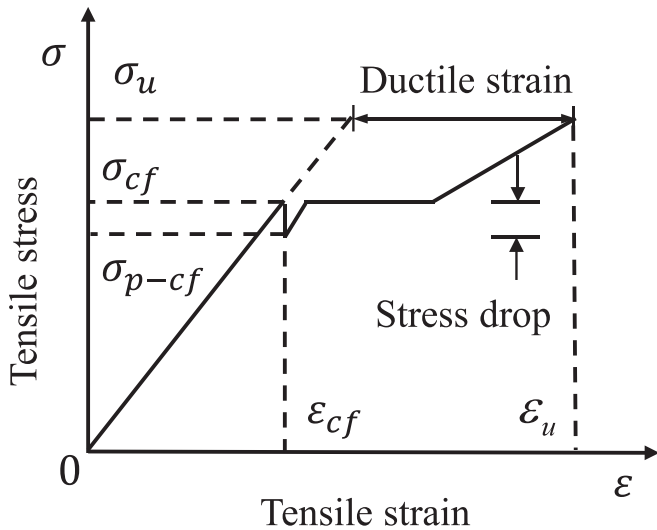


Fig. 14. Ductile strain and stress drop.

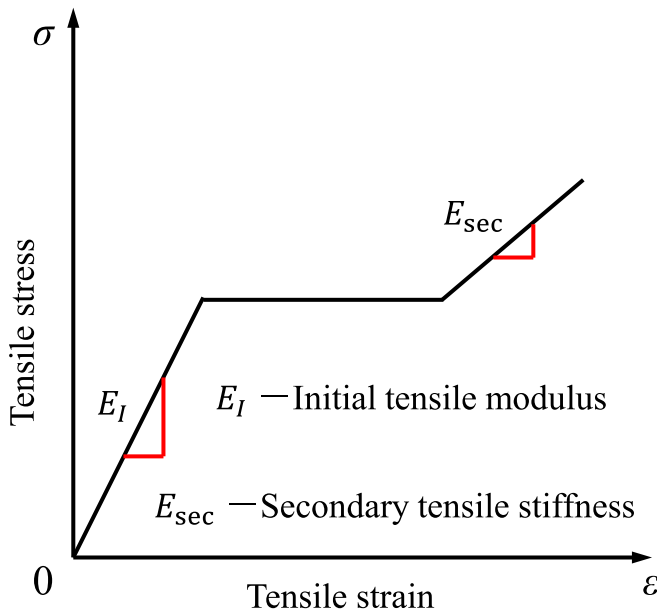


Fig. 15. Diagram of secondary stiffness of C/G-HFRP bar.

Based on the experiment results in this study,  $\alpha$  can be determined as follows:

$$\alpha = \frac{E_{sec}^E - E_M(1 - V_f)}{V_f V_C E_C} - \frac{V_G E_G}{V_C E_C} \quad (12)$$

where  $E_{sec}^E$  is the experimental secondary stiffness of HFRP bar, which can be determined by averaging the slopes of straight lines at the strain hardening stages in the tensile stress-strain response of each group of C/G-HFRP bars.

By substituting the parameters in Table 2 and the experimental secondary stiffness into Eq. (12),  $\alpha$  can be determined, as shown in Table 7. The average value of  $\alpha$  was  $-0.016$  in this study. The small value of  $\alpha$  indicates that the contribution of failed carbon fibers to the secondary stiffness in the tensile stress-strain response of C/G-HFRP bars can be omitted.

Table 7

Experimental secondary stiffness of C/G-HFRP bars after modification.

| Sample code | $E_{sec}^E$ (GPa) | $\alpha$ |
|-------------|-------------------|----------|
| HC1G10C60   | $46.8 \pm 1.51$   | $-0.061$ |
| HC1G8C60    | $44.3 \pm 1.11$   | $-0.032$ |
| HC1G6C60    | $43.2 \pm 2.77$   | $0.031$  |
| HC1G4C60    | —                 | —        |
| HC1G10D60   | $47.2 \pm 1.25$   | $-0.040$ |
| HC1G8D60    | $45.1 \pm 1.02$   | $0.001$  |
| HC1G6D60    | $42.1 \pm 1.82$   | $-0.005$ |
| HC1G4D60    | —                 | —        |
| HC1G6C50    | $36.2 \pm 3.47$   | $-0.018$ |
| HC1G6C70    | $47.5 \pm 1.61$   | $-0.008$ |
| Average     |                   | $-0.016$ |

## 6. Conclusions

In this work, 0.3 wt% graphene nanosheets (GNPs) was incorporated into the polymer matrix to prepare the carbon/glass hybrid fiber-reinforced GNPs-modified polymer (C/G-HFRP) bars, while the corresponding C/G-HFRP bars without modification using GNPs were also prepared for comparison. The effects of GNPs-modified polymer matrix on the tensile properties of C/G-HFRP bars were investigated. The following conclusions can be drawn.

- (1) The addition of GNPs improves the fracture toughness of the polymer matrix between fibers to resist the energy release rate of C/G-HFRP bars at the failure of carbon fibers, which reduces the separation degree of glass fibers around the failed carbon fibers.
- (2) The tensile strength and strain of the modified C/G-HFRP bars at the initial failure of carbon fibers and at the ultimate state are averagely improved by 6.9%, 4.3%, 7.1% and 7.1% relative to those of the unmodified C/G-HFRP bars, respectively.
- (3) A relative complete tensile stress-strain response for the modified C/G-HFRP bars can be experimentally obtained by strain gauge data, which benefits from the global structure of glass fibers around the failed carbon fibers. The multi-stage stress-strain responses confirm the ductile failure of C/G-HFRP bars under tensile load.
- (4) The hybrid effect, ductile strain, stress drop rate, and strength ratio can be used as the evaluation indicators to reflect the superior properties of modified C/G-HFRP bars under tensile load.
- (5) The contribution of carbon fibers to the secondary stiffness in the tensile stress-strain response of C/G-HFRP bars can be omitted.

## CRediT authorship contribution statement

**Yu Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Danying Gao:** Writing – review & editing, Resources, Project administration, Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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